Taylor & Francis
Taylor & Francis Group

Annals of Tropical Medicine & Parasitology

ISSN: 0003-4983 (Print) 1364-8594 (Online) Journal homepage: http://www.tandfonline.com/loi/ypgh19

Transmission and epidemiology of newly transmitted falciparum malaria in the central highland plateaux of Madagascar

J. P. Lepers, D. Fontenille, M. D. Rason, C. Chougnet, P. Astagneau, P. Coulanges & P. Deloron

To cite this article: J. P. Lepers, D. Fontenille, M. D. Rason, C. Chougnet, P. Astagneau, P. Coulanges & P. Deloron (1991) Transmission and epidemiology of newly transmitted falciparum malaria in the central highland plateaux of Madagascar, Annals of Tropical Medicine & Parasitology, 85:3, 297-304, DOI: 10.1080/00034983.1991.11812564

To link to this article: http://dx.doi.org/10.1080/00034983.1991.11812564

Published online: 15 Nov 2016.

Submit your article to this journal

View related articles

Citing articles: 8 View citing articles

Full Terms & Conditions of access and use can be found at
http://www.tandfonline.com/action/journalInformation?journalCode=ypgh19

Download by: [Australian Catholic University] **Date:** 22 August 2017, At: 04:43

Annals of Tropical Medicine and Parasitology, Vol. 85, No. 3, 297–304 (1991)

Transmission and epidemiology of newly transmitted falciparum malaria in the central highland plateaux of Madagascar

By J. P. LEPERS*, D. FONTENILLE*, M. D. RASON*, C. CHOUGNET†, P. ASTAGNEAU†, P. COULANGES* AND P. DELORON†‡§

**Unité de Recherche sur le Paludisme, Institut Pasteur de Madagascar, B.P. 1274, Antananarivo, Madagascar*
†INSERM Unite 13, Hôpital Claude Bernard, 10 Avenue de la Porte d'Aubervilliers, 75944 Paris Cedex 19, France
‡Institut de Médecine et d'Epidémiologie Africaines (IMEA), Hôpital Claude Bernard, France

Received 13 July 1990, Accepted 8 September 1990

Downloaded by [Australian Catholic University] at 04:43 22 August 2017

The recent reappearance of *Plasmodium falciparum* in the central highland plateaux of Madagascar has led to an important increase in both morbidity and mortality in the population. To understand the phenomena that originated this outbreak, and to evaluate the clinical and biological reactions of the population, we conducted entomological surveys during the whole malaria transmission season of 1988. In parallel, 41 individuals presenting with a malaria attack in January were given a curative antimalarial treatment and were followed weekly for 20 weeks, until the end of the transmission season, in June. During the follow-up, individuals presenting with a clinical malaria attack or with a parasite density above 5000 per mm^3 of blood were again given antimalarial treatment. Overall, individuals presented with an average of 5·6 thick blood smears positive for *P. falciparum*, and 1·4 malaria attacks requiring treatment. *Plasmodium falciparum* prevalence rates gradually increased to 53·8% in April, one to two months after the observed peak of transmission, then decreased to 15·4% at the end of follow-up. The proportion of *P. falciparum* infections with clinical symptoms did not follow a similar temporal evolution and was significantly lower than at enrolment. Splenic rates gradually decreased from 29·3–2·6%, while anti-*P. falciparum* and anti-Pf155/RESA antibodies increased. For the whole transmission season individuals experienced a mean 1·5 infective bites; the malaria vectors were *Anopheles gambiae* s.l and *An. funestus*. This study demonstrates that, in this population newly exposed to *P. falciparum* infections, almost all sporozoite inoculations lead to blood parasite infection requiring treatment. Early therapy of malaria attacks can reduce malaria-related morbidity, without impairing the development of an antimalarial immune response.

Falciparum malaria has recently reappeared in the central highland plateaux of Madagascar. Intensive control activities conducted in the early 1950s had led to the disappearance of *Anopheles funestus*, and to malaria eradication in nearly the whole plateaux area. In 1986 *An. funestus* was shown to be present and to act as a malaria vector (Fontenille and Rakotoarivony, 1988). In early 1987, a sudden increase in the morbidity and the mortality in the population of the central plateaux was demonstrated to be due to *Plasmodium falciparum* malaria (Lepers *et al.*,

§Address for correspondence and reprint requests: P. Deloron, INSERM U13, Hôpital Claude Bernard, 10 Avenue de la Porte d'Aubervilliers, 75944 PARIS Cedex 19, France.

0003–4983/91/030297+08 $03.00/0

© 1991 Liverpool School of Tropical Medicine

Downloaded by [Australian Catholic University] at 04:43 22 August 2017

1988). In May 1987 we established a field health-centre in Manarintsoa, a small village in the middle of the outbreak area. By treating all malaria cases with a standard chloroquine regimen we were able to reduce mortality to its previous level. Since then malaria seems to be highly endemic, *P. falciparum* being by far the species most encountered (Lepers *et al.*, 1989*a*).

To understand the phenomena that produced this outbreak, and to evaluate the clinical and biological reactions of the population, we conducted entomological surveys as well as a cohort study during the whole transmission season of 1987–1988.

SUBJECTS AND METHODS

Study Area

The study took place in Manarintsoa, a village with about 1550 inhabitants, located in the central highland plateaux, 20 km west of Antananarivo, the capital city of Madagascar. The village is divided into about 30 hamlets, at an elevation of 1200–1400 m, with flooded rice fields. A cold and dry season from June to November, with average nightly temperatures between 10° and 15°C, is unfavourable to the development of the mosquito malaria vectors. A hot and humid season from late November to May, with average temperatures at night between 15° and 18°C, allows the multiplication of *An. gambiae* s.l. and *An. funestus*, which are the only known malaria vectors of the area. Schistosomiasis is not endemic in this area of Madagascar.

Entomological Methods

Entomological surveys were conducted monthly from October 1987 to July 1988. Culicidae were caught from 18.00–5.00 hours on human bait, by three individuals outdoors and three individuals indoors. Endophilic mosquitoes were collected in 50 bedrooms. *Anopheles* specimens were dissected for sporozoite detection and ovarian age determination. Selected mosquitoes were reacted by ELISA using anticircumsporozoite protein monoclonal antibody against each of the four human malaria parasite species, as described by Burkot *et al.* (1984) with minor modifications.

Cohort Study

SUBJECTS INVESTIGATED

At the very beginning of the transmission season, during a two-week period in early January 1988, all subjects more than 10 years old and presenting to the health centre of Manarintsoa were submitted to a clinical examination and a thick blood smear was taken. Individuals with a malaria attack were enrolled according to the following criteria: (a) to present with a pure *P. falciparum* parasitaemia with a density above 1000 mm^{-3} of blood; (b) to be permanent resident of the village for at least three years; and (c) to give informed consent to participate to the study. Individuals with mixed *Plasmodium* infection or with other febrile illness were excluded. Urines were collected to perform a chloroquine urine test. Venous blood was drawn from enrolled individuals for serum collection, and the thick blood smear was repeated. The subjects were then given a three-day oral quinine-based regimen as a curative treatment of their infection (Deloron *et al.*, 1989).

FOLLOW-UP

Every week for 20 weeks, i.e. until the end of the malaria transmission season, each subject was submitted to a clinical examination, a chloroquine urine test, and blood collection for thick and thin blood smear examination. Every two weeks, venous blood was drawn for serum collection. To identify every malaria attack episode occurring in the cohort, a physician lived permanently in Manarintsoa and was available at any time to examine each subject complaining of any illness. Treatment was not administered before parasites were present in the blood, but only in subjects presenting with a clinical malaria attack or with a parasite density above 5000 mm^{-3} of blood, who were given a three-day WHO standard regimen of chloroquine (25 mg kg^{-1} over three days). As numerous chloroquine-resistant strains are present in this area (Lepers *et al.*, 1989*b*), a second treatment with amodiaquine (25 mg kg^{-1} over three days) was given when parasites were present seven days after chloroquine administration.

Downloaded by [Australian Catholic University] at 04:43 22 August 2017

LABORATORY METHODS

Blood smears were Giemsa-stained and searched for malaria parasites by examination against 1000 leucocytes. Parasite densities were calculated from the number of asexual parasites observed, assuming an average leucocyte count of 8000 mm^{-3} of blood.

Antimalarial drug consumption was assessed by the Bergqvist urine test for 4-aminoquinolines (Bergqvist *et al.*, 1985).

Antibodies against total *P. falciparum* asexual parasites were measured by a conventional immunofluorescent assay (IFA) using a thick blood smear of *P. falciparum*-infected erythrocytes as antigen (threshold of positivity: a reciprocal titre of 64) (Sulzer *et al.*, 1969). Antibodies to Pf155/RESA, a putative protective antigen (Collins *et al.*, 1986; Perlmann *et al.*, 1987), were measured by erythrocyte membrane immunofluorescence (EMIF) carried out on a gluteraldehyde-fixed monolayer of *P. falciparum* ring-infected erythrocytes, as previously described (threshold of positivity: a reciprocal titre of 4) (Perlmann *et al.*, 1984). Both immunofluorescent techniques were carried out with a *P. falciparum* strain isolated in Manarintsoa.

STATISTICAL ANALYSIS

Differences in proportions were tested by the Chi-square (χ^2) test and by the χ^2 test for trend. Differences in means were tested by Student's *t*-test. *P* values $<0{\cdot}05$ were considered significant.

RESULTS

Entomological Surveys

A total of 16 163 Culicidae specimens were collected on human bait during 294 man-nights, including 311 *An. gambiae* s.l. and 11 *An. funestus*. One additional *An. gambiae* s.l. and six *An. funestus* specimens were collected among indoor-resting mosquitoes. Fifty-seven per cent of bites by *An. gambiae* s.l. occurred before 22.00 hours, probably as a consequence of the coolness of the nights in the plateaux area. The highest aggressiveness was observed from January to May (Fig. 1). Overall, 76% of *An. gambiae* fed outdoors.

An average of 51% of the *An. gambiae* s.l. specimens were parous. Presence of *Plasmodium* sporozoites in the salivary glands was demonstrated in one *An. gambiae* s.l. (caught on man, outdoors at 22.00 hours in late February) and one *An. funestus* specimen (caught among indoor-resting mosquitoes, in late February). Dissection of the salivary glands of 278 *An. coustani* and 156 *An. squamosus* or *An. cydippis* failed to identify sporozoites in any individual. Similarly, no positive reaction was obtained by reacting 445 *An. coustani* and 720 *An. squamosus* or *An. cydippis* in ELISA using anticircumsporozoite protein monoclonal antibody. From November to May, during the rainy season, the sporozoite index(s) was 0·71% for both *An. gambiae* s.l. and *An. funestus*. The number of bites by malaria vector per man per night (ma) was 1·3. For the whole transmission season, the number of infective bites per man can be calculated as 1·5 and, according to Krafsur and Armstrong (1978), the infection risk as 0·86 (the daily risk being 0·0092).

Cohort Study

Forty-one patients were enrolled and successfully followed during the whole transmission season; 17 were male and 24 were female. The mean age ($\pm$S.D.) of the cohort was 25·3 ($\pm$11·9) years. Among these 41 individuals, 21 (61%) were febrile (axillary temperature $>37{\cdot}5$°C) at enrolment, and 22 (53·7%) presented with other malaria symptoms: headache, vomitus, or diarrhoea. None suffered from a severe malaria attack. Geometric mean parasite density (GMPD) was 7183 parasites mm^{-3} of blood, and 18 (43·9%) had a positive chloroquine urine test. Splenic rate was 29·3%.

The number of subjects attending weekly examinations ranged from 35–41 (mean attendance 38·8), representing a total of 772 follow-up consultations. A total of 751 blood smears were obtained, of which 255 (33%) were positive for malaria parasites. *Plasmodium falciparum* was identified in 226 (88·6%) slides, and *P. vivax* or *P. ovale* in 38 (15%) slides; nine slides showed a mixed infection. Following the initial treatment all individuals were clear from parasitaemia, and throughout week 1 (W1) thick smears remained negative for malaria parasites. During W2 five

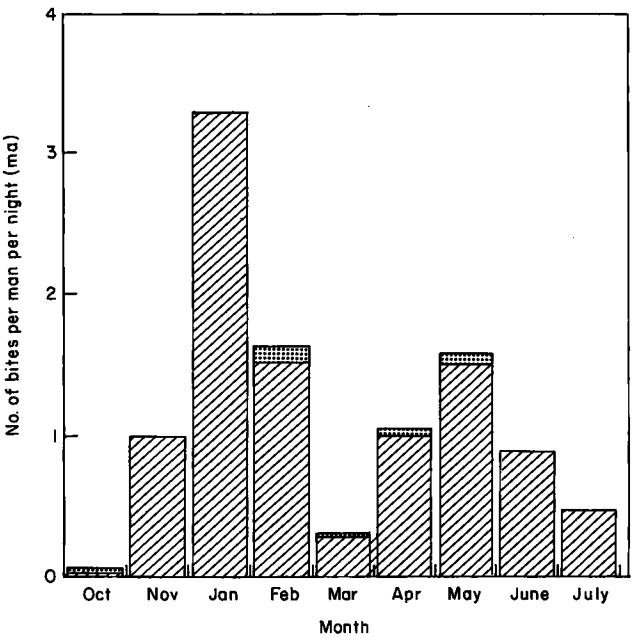


Fig. 1. Number of bites per man per night by *Anopheles gambiae* s.l. (▨) and *An. funestus* (▩) occurring in Manarintsoa from October 1987 to July 1988, by month. No survey was conducted in December 1987.

subjects presented with *P. falciparum* parasites in their blood. Afterwards, *P. falciparum* prevalence rates increased gradually to a peak value of 53·8% in April, during W12, then decreased to 15·4% during W20, at the end of the follow-up period (Fig. 2). Overall, individuals presented with an average of $5{\cdot}6 \pm 3{\cdot}1$ (range 0–13) thick blood smears positive for *P. falciparum* during the 20-weeks follow up. Monthly incidence was 24% in February, 38% in March, 36% in April, 54% in May, and 17% in June. Among *P. falciparum*-infected individuals, GMPDs decreased rapidly to average 2000 parasites mm^{-3} of blood after W4.

During the follow up, the proportion of *P. falciparum*-infected individuals who experienced clinical symptoms of malaria (20·8%) was lower than at enrolment (χ^2 test, $P < 0{\cdot}001$). Although the highest proportion of such individuals was observed in W12, there was no evidence of any tendency for this proportion to increase with time (χ^2 test for trend, not significant). Similarly, the proportion of malaria attacks requiring treatment was also highly variable from one week to another. Twenty-six of the 41 (63·4%) individuals presented with one or more treated malaria attacks during the follow-up. Overall, individuals presented with an average of $1{\cdot}4 \pm 1{\cdot}4$ (range 0–4) malaria attacks requiring treatment. Nine among the 56 treated malaria attacks (16%) had parasite densities over 5000 mm^{-3} of blood without any clinical symptoms.

A total of 610 urine samples were obtained from the 772 (79%) follow-up visits. Of those, 111 (17·3%, varying weekly from 4–29·7%) produced a positive reaction in the Bergqvist urine test. However, almost half (46·8%) of the positive reactions were observed in individuals whom we had treated for malaria in the previous two weeks. Consequently, only 59 (9·2%) urine tests demonstrated an uncontrolled chloroquine consumption. *Plasmodium falciparum* prevalence rates were higher in

Downloaded by [Australian Catholic University] at 04:43 22 August 2017

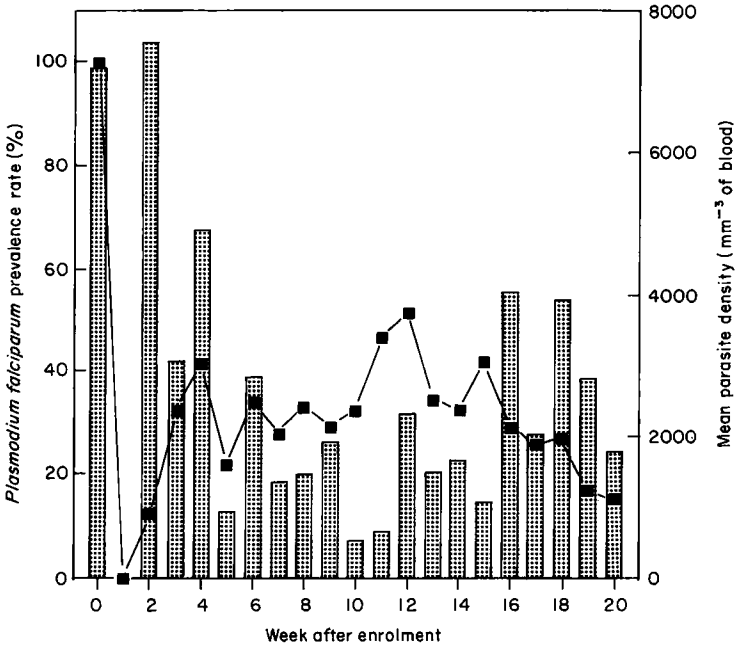


Fig. 2. Weekly evolution of *Plasmodium falciparum* prevalence rates (bars) and mean parasite densities (line) in a cohort of inhabitants of Manarintsoa, Madagascar.

individuals presenting with a non-therapeutic positive chloroquine urine test than in those with a negative test (51% and 29·3%, respectively) (Chi square test, $P < 10^{-6}$). However, among *P. falciparum*-infected individuals presenting with or without a positive chloroquine urine test, similar GMPDs (2850, 95% C.I.: 33–245 471; and 1884, 95% C.I.: 12–293 089, respectively) (Student's *t*-test, N.S.) and malaria attack rates (28·1% and 30·4%, respectively) (χ^2 test, N.S.) were observed.

Splenic rates gradually decreased from 29·3% at enrolment to 2·6% at W20 (Chi-square test for trend, $P < 0{\cdot}01$) (Table). Conversely, anti-Pf155/RESA antibody positivity rates, as detected by EMIF, increased from 25–50% during the follow up (χ^2 test for trend, $P < 10^{-3}$). Levels of anti-*P. falciparum* and anti-Pf155/RESA antibodies also increased during the follow-up, with geometrical mean reciprocal titres (GMRT) varying from 1097–2702, and from 2·5–6, respectively.

DISCUSSION

This study, confirming our previous work (Lepers *et al.*, 1989*a*), demonstrates that falciparum malaria is currently highly endemic in this area of Madagascar. Although we treated all *P. falciparum* infections with clinical symptoms or high parasite counts occurring in this cohort of adults, monthly incidence rates of *P. falciparum* infection were consistently above 24% from February to May. Peak prevalence was observed in April, following the malaria vectors biting peak occurring in January–February. Transmission of the disease is by *An. funestus* and *An. gambiae* s.l. Further studies, as well as the results of Ralisoa Randrianasolo and Coluzzi (1987), demonstrate that *An. arabiensis* is probably the only species in the gambiae complex present in this area. Only two mosquito specimens with sporozoites in the salivary glands were caught, both at the end of February, evidencing the low level of the transmission,

Downloaded by [Australian Catholic University] at 04:43 22 August 2017

TABLE

Development of spleen rates and antibody responses to total Plasmodium falciparum *parasites (IFA) and to Pf155/RESA (EMIF) in a cohort of inhabitants of Manarintsoa, Madagascar*

	Week after enrolment					
	0	*4*	*8*	*12*	*16*	*20*
Spleen rate (%)	*29·3*	*20*	*17·9*	*10·3*	*8·3*	*2·6*
IFA %+	97·5	100	100	100	100	100
GMRT*	1097	1755	2012	1796	2274	2702
EMIF %+	25	36·4	38·5	43·2	54·5	50
GMRT*	2·5	4	4·6	5·2	6·1	5·9

*Geometric mean reciprocal titre.

with about 1·5 infective bites per man per year. The individual inoculation risk is near 0·86 for the total duration of the study. During the whole transmission season, individuals presented with an average of 1·4 malaria attacks that necessitated therapy. Although some of these malaria attacks were probably due to the late recrudescence of a chloroquine-resistant parasite, the low sporozoite inoculation rate occurring in this area of Madagascar indicates that almost all infected bites are followed by subsequent parasite blood infection that will have to be treated. This also demonstrates the low level of immune protection in this population newly exposed to falciparum malaria.

We previously reported data collected the same year from out-patient clinic consultants of the same village (Lepers *et al.*, 1990). In this study, after the beginning of the transmission season, parasite rates and parasite densities increased (from 45–75%, and from 8682–11 193 parasites mm^{-3} of blood, respectively), while febrile illnesses associated with malaria were decreasing (from 71–42·5%). During the second part of the rainy season parasite densities began to decrease (to 3810 parasites mm^{-3} of blood) while parasite rates and malaria-associated fevers remained at the same level. In the present study of a closely monitored cohort, *P. falciparum* prevalence rates and malaria-associated fever rates were lower. No seasonal variations in the clinical impact of *P. falciparum* malaria were observed. Individuals were examined for malaria parasites every week. Therapy was initiated when clinical symptoms were present, but also when parasite counts were high (above 5000 mm^{-3} of blood), to reduce the possibility of the subsequent appearance of illness (Greenwood *et al.*, 1987). This observation emphasizes the fact that malaria morbidity is reduced by early therapy, even in this context of newly transmitted chloroquine-resistant falciparum malaria. In addition, the decrease of splenic rates, as well as the increase of both anti-*P. falciparum* and anti-Pf155/RESA antibodies, during the malaria transmission season demonstrate that such strategy does not impair the development of the immune response (Baudon *et al.*, 1986).

Uncontrolled drug consumption was observed in 9% of urine samples. Individuals with previous intake of chloroquine were more likely to present with *P. falciparum* than individuals that did not take this drug, but malaria attack rates were similar in both groups. Whether this apparent lack of action of chloroquine is related to an insufficient prophylatic or therapeutic drug dosage, to parasite resistance to chloroquine, or to the fact that subjects with higher parasite densities were more likely to be sick and to take antimalarial drugs than the other individuals, is questionable. However, the permanent presence of a physician of our team in the

Downloaded by [Australian Catholic University] at 04:43 22 August 2017

village should have motivated individuals feeling sick to search for free medical advice and therapy.

In conclusion, despite a very low level of transmission, *P. falciparum* malaria is highly endemic in the central highland plateaux of Madagascar. Malaria-related morbidity may be greatly reduced by adequate therapy. Although numerous chloroquine-resistant strains of *P. falciparum* are present in this area, 4-aminoquinolines appear to be still able to induce clinical improvement of patients. In this population newly exposed to malaria, the rapid development of the immune response (and in particular of anti-Pf155/RESA antibody, although only to a low titre) is somewhat reassuring, and might indicate that malaria-related illnesses will be less prevalent during subsequent transmission seasons.

ACKNOWLEDGEMENTS. This work was supported by grants from AUPELF/UREF and from the Tropical Pathology Society. We thank Gary H. Campbell, of the Malaria Branch of the Centers for Disease Control, for the kind supply of anticircumsporozoite protein monoclonal antibodies. We also thank the population of Manarintsoa for their collaboration.

REFERENCES

BAUDON, D., ROUX, J., CARNEVALE, P., REY, J. L., MEYRAN, M. B. & BRANDICOURT, O. (1986). Systematic chemotherapy of febrile cases: a substitute strategy for malaria control in rural areas of Africa. *Transactions of the Royal Society of Tropical Medicine and Hygiene*, **80**, 164.

BERGQVIST, Y., HED, C., FUNDING, L. & SUTHER, A. (1985). Determination of chloroquine and its metabolites in urine: a field method based on ion-pair extraction. *Bulletin of the World Health Organization*, **63**, 893–898.

BURKOT, T. R., WILLIAMS, J. L. & SCHNEIDER, I. (1984). Identification of *Plasmodium falciparum*-infected mosquitoes by a double antibody enzyme-linked immunosorbent assay. *American Journal of Tropical Medicine and Hygiene*, **33**, 783–788.

COLLINS, W. E., ANDERS, R. F., PAPPAIOANOU, M. CAMPBELL, G. H., BROWN, G. V., KEMP, D. J., COPPEL, R. L., SKINNER, J. C., ANDRISYAK, P. M., FAVAROLO, J. M., CORCORAN, L. M., BRODERSON, J. R., MITCHELL, G. F. & CAMPBELL, C. C. (1986). Immunization of *Aotus* monkeys with recombinant proteins of an erythrocyte surface antigen of *Plasmodium falciparum*. *Nature*, **323**, 259–262.

DELORON, P., LEPERS, J. P., VERDIER, F., CHOUGNET, C. RAMANAMIRIJA, J. A., ANDRIAMANGATIANA-RASON, M. D., COULANGES, P. & JAUREGUIBERRY, G. (1989). Efficacy of a 3-day oral regimen of a quinine-quinidine-cinchonine association (Quinimax®) for treatment of falciparum malaria in Madagascar. *Transactions of the Royal Society of Tropical Medicine and Hygiene*, **83**, 751–754.

FONTENILLE, D. & RAKOTOARIVONY, I. (1988). Reappearance of *Anopheles funestus* as a malaria vector in the Antananarivo region, Madagascar. *Transactions of the Royal Society of Tropical Medicine and Hygiene*, **82**, 644–645.

GREENWOOD, B. M., BRADLEY, A. K., GREENWOOD, A. M., BYASS, P., JAMMEH, K., MARSH, K. TULLOCH, S., OLDFIELD, F. S. J. & HAYES, R. (1987). Mortality and morbidity from malaria among children in a rural area of The Gambia, West Africa. *Transactions of the Royal Society of Tropical Medicine and Hygiene*, **81**, 478–486.

KRAFSUR, E. S. & AMSTRONG, J. C. (1978). An integrated view of entomological and parasitological observations of falciparum malaria in Gambela, Western Ethiopian Lowlands. *Transactions of the Royal Society of Tropical Medicine and Hygiene*, **72**, 348–356.

LEPERS, J. P., ANDRIAMANGATIANA-RASON, M. D., RAMANAMIRIJA, J. A., FONTENILLE, D., LEPERS, C., DELORON, P. & COULANGES, P. (1989*a*). Le paludisme en 1988 dans un village des Hauts Plateaux malgaches. Données épidémiologiques. *Archives de l'Institut Pasteur de Madagascar*, **56**, 97–130.

LEPERS, J. P., DELORON, P., MOUDEN, J. C., LE BRAS, J. & COULANGES, P. (1989*b*). Sudden increase in number of isolates of *Plasmodium falciparum* resistant to chloroquine in Madagascar. *Transactions of the Royal Society of Tropical Medicine and Hygiene*, **83**, 491–492.

LEPERS, J. P., DELORON, P., ANDRIAMANGATIANA-RASON, M. D., RAMANAMIRIJA, J. A. & COULANGES, P. (1990). Newly transmitted *Plasmodium falciparum* malaria in the central highland plateaux of

Downloaded by [Australian Catholic University] at 04:43 22 August 2017

Madagascar: assessment of its clinical impact in a rural community. *Bulletin of the World Health Organization,* **68,** 217–222.

LEPERS, J. P., DELORON, P., FONTENILLE, D. & COULANGES, P. (1988). Reappearance of falciparum malaria in Central Highland Plateaux of Madagascar. *Lancet,* **i,** 586.

PERLMANN, P., BERZINS, K., KABILAN, L., PERLMANN, H., TROYE-BLOMBERG, M., WAHLIN, B., BJÖRKMAN, A., HÖGH, B., PETERSEN, E. & PATARROYO, E. M. (1987). Humoral and cellular immune responses to the carboxy-terminal amino acid repeats of Pf155, a potential vaccine candidate of the human malaria parasite *Plasmodium falciparum.* In *Vaccines 87,* eds. Chanock, R. M., Lerner, R. A., Brown, F. & Ginsberg, H. pp. 86–91. New York: Cold Spring Harbor Laboratory.

PERLMANN, H., BERZINS, K., WALGREN, M., CARLSSON, J., BJÖRKMAN, A., PATARROYO, M. E. & PERLMANN, P. (1984). Antibodies in malarial sera to parasite antigens in the membrane of erythrocytes infected with early asexual stages of *Plasmodium falciparum. Journal of Experimental Medicine,* **159,** 1686–1704.

RALISOA RANDRIANASOLO, B. O. & COLUZZI, M. (1987). Genetical investigations on zoophilic and exophilic *Anopheles arabiensis* from Antananarivo area (Madagascar). *Parassitologia,* **29,** 93–97.

SULZER, A., WILSON, M. & HALL, E. C. (1969). Indirect fluorescent-antibody tests for parasitic diseases. An evaluation of a thick-smear antigen in the IFA test for malaria antibodies. *American Journal of Tropical Medicine and Hygiene,* **18,** 199–205.

Downloaded by [Australian Catholic University] at 04:43 22 August 2017